SCUT Data Analysis And Modeling

Lab 6: Inference for numerical data

Use the template

Before you begin the lab, don't forget to use the template used in the previous lab lesson.

In this lab we'll investigate the probability distribution that is most central to statistics: the normal distribution. If we are confident that our data are nearly normal, that opens the door to many powerful statistical methods. Here we'll use the graphical tools of R to assess the normality of our data and also learn how to generate random numbers from a normal distribution.

North Carolina births

In 2004, the state of North Carolina released a large data set containing information on births recorded in this state. This data set is useful to researchers studying the relation between habits and practices of expectant mothers and the birth of their children. We will work with a random sample of observations from this data set.

Exploratory analysis

Load the `nc` data set into our workspace.

```
load(url("http://www.stat.duke.edu/~cr173/Sta102_Sp16/Lab/nc.RData"))
```

We have observations on 13 different variables, some categorical and some numerical. The meaning of each variable is as follows.

variable	description
`fage`	father's age in years.
`mage`	mother's age in years.
`mature`	maturity status of mother.
`weeks`	length of pregnancy in weeks.
`premie`	whether the birth was classified as premature (premie) or full-term.
`visits`	number of hospital visits during pregnancy.
`marital`	whether mother is `married` or `not married` at birth.
`gained`	weight gained by mother during pregnancy in pounds.

variable	description
`weight`	weight of the baby at birth in pounds.
`lowbirthweight`	whether baby was classified as low birthweight (`low`) or not (`not low`).
`gender`	gender of the baby, `female` or `male` .
`habit`	status of the mother as a `nonsmoker` or a `smoker` .
`whitemom`	whether mom is `white` or `not white` .

Exercise 1 What are the cases in this data set? How many cases are there in our sample?

As a first step in the analysis, we should consider summaries of the data. This can be done using the `summary` command:

```
summary(nc)
```

As you review the variable summaries, consider which variables are categorical and which are numerical. For numerical variables, are there outliers? If you aren't sure or want to take a closer look at the data, make a graph.

Consider the possible relationship between a mother's smoking habit and the weight of her baby. Plotting the data is a useful first step because it helps us quickly visualize trends, identify strong associations, and develop research questions.

Exercise 2 Make a side-by-side boxplot of `habit` and `weight` . What does the plot highlight about the relationship between these two variables?

The box plots show how the medians of the two distributions compare, but we can also compare the means of the distributions using the following function to split the `weight` variable into the `habit` groups, then take the mean of each using the `mean` function.

```
by(nc$weight, nc$habit, mean)
```

There is an observed difference, but is this difference statistically significant? In order to answer this question we will conduct a hypothesis test .

Inference

Exercise 3 Write the hypotheses for testing if the average weights of babies born to smoking and non-smoking mothers are different.

Next, we introduce a new function, `inference` , that we will use for conducting hypothesis tests and constructing confidence intervals. First, load the function:

```
load(url("http://www.stat.duke.edu/~cr173/Sta102_Sp16/Lab/inference.RData"))
```

Then, run the following:

```
inference(y = nc$weight, x = nc$habit, est = "mean", type = "ht", null = 0, alternative = "twosided",
method = "theoretical")
```

Let's pause for a moment to go through the arguments of this custom function. The first argument is `y`, which is the response variable that we are interested in: `nc$weight`. The second argument is the explanatory variable, `x`, which is the variable that splits the data into two groups, smokers and non-smokers: `nc$habit`. The third argument, `est`, is the parameter we're interested in: `"mean"` (other options are `"median"`, or `"proportion"`.) Next we decide on the `type` of inference we want: a hypothesis test (`"ht"`) or a confidence interval (`"ci"`). When performing a hypothesis test, we also need to supply the `null` value, which in this case is `0`, since the null hypothesis sets the two population means equal to each other. The `alternative` hypothesis can be `"less"`, `"greater"`, or `"twosided"`. Lastly, the `method` of inference can be `"theoretical"` or `"simulation"` based.

Note that the `inference` function you are introduced to in this lab will be very useful for your projects. More information on the function, as well as many other functions commonly used in Sta 101, can be found at https://stat.duke.edu/~mc301/R/Rcommands.html (https://stat.duke.edu/%7Emc301/R/Rcommands.html).

Exercise 4 Change the `type` argument to `"ci"` to construct and record a confidence interval for the difference between the weights of babies born to smoking and non-smoking mothers. Interpret this interval.

By default the function reports an interval for ($\mu_{nonsmoker} - \mu_{smoker}$). We can easily change this order by using the `order` argument:

```
inference(y = nc$weight, x = nc$habit, est = "mean", type = "ci", null = 0, alternative = "twosided",
method = "theoretical", order = c("smoker","nonsmoker"))
```

Additional Exercises

Exercise 5 Calculate a 95% confidence interval for the average length of pregnancies (`weeks`) and interpret it in context. Note that since you're doing inference on a single population parameter, there is no explanatory variable, so you can omit the `x` variable from the function.

Exercise 6 Would you expect a 90% confidence interval to be wider or narrower? Confirm by creating the confidence interval for the same parameter at the 90% confidence level. You can change the confidence level by adding a new argument to the function: `conflevel = 0.90`.

Exercise 7 Conduct a hypothesis test evaluating whether the average weight gained by younger mothers is different than the average weight gained by mature mothers. . Interpret the p-value. Make sure that your interpretation is in context of the data.

Exercise 8 Now, a non-inference task: Determine the age cutoff for younger and mature mothers. Use a method of your choice, and explain how your method works.

Exercise 9 Pick a pair of numerical and categorical variables and come up with a research question evaluating the relationship between these variables. Formulate the question in a way that it can be answered using a hypothesis test and/or a confidence interval. Answer your question using the `inference` function, report the statistical results, and also provide an explanation in plain language

Submission

When you are done, regarding the submission of the lab report, please note the following:

1. The report needs to include Exercise1-9;

2. Some exercises may not need to enter the code, please enter your text to express your understanding;

3. Please enter your student IDs and names of your group at the top of the report;

4. Please submit the lab report within 24 hours after class;

5. The submitted code must be able to run successfully at one time;

6. Please complete the lab report **in HTML and RMD formats, package the file into a .zip compressed file and upload it to the Blackboard learning (BB) system**, and for each group only one lab report is needed;

7. Please name your file following the format: "Student ID 1-Name 1-Student ID 2-Name 2_Lab Number_Course Name". For example, 2020XXXXXXXX-张三-2020XXXXXXXX-李四_Lab1_Data-Analysis-and-Modeling.